

CE/CZ4055 Cyber Physical System Security

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CCDS, NTU



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CPSSECURITY



Contents

➔ CPS/IoT: Again

- Smart Cards
- Discussion



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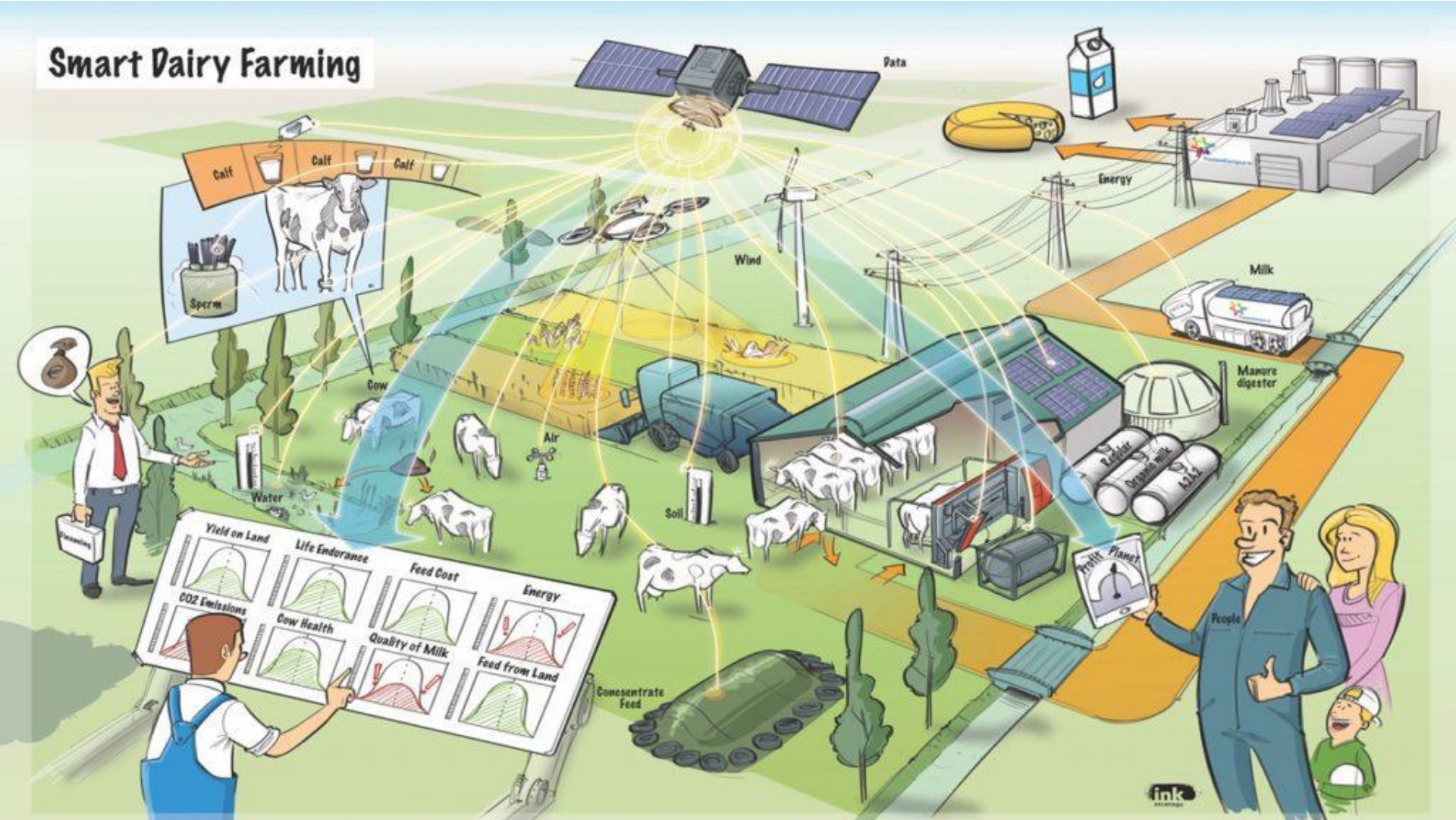
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Smart Dairy Farming





IOT in Agriculture



DreamzTech Solutions



NANYANG
TECHNOLOGICAL
UNIVERSITY

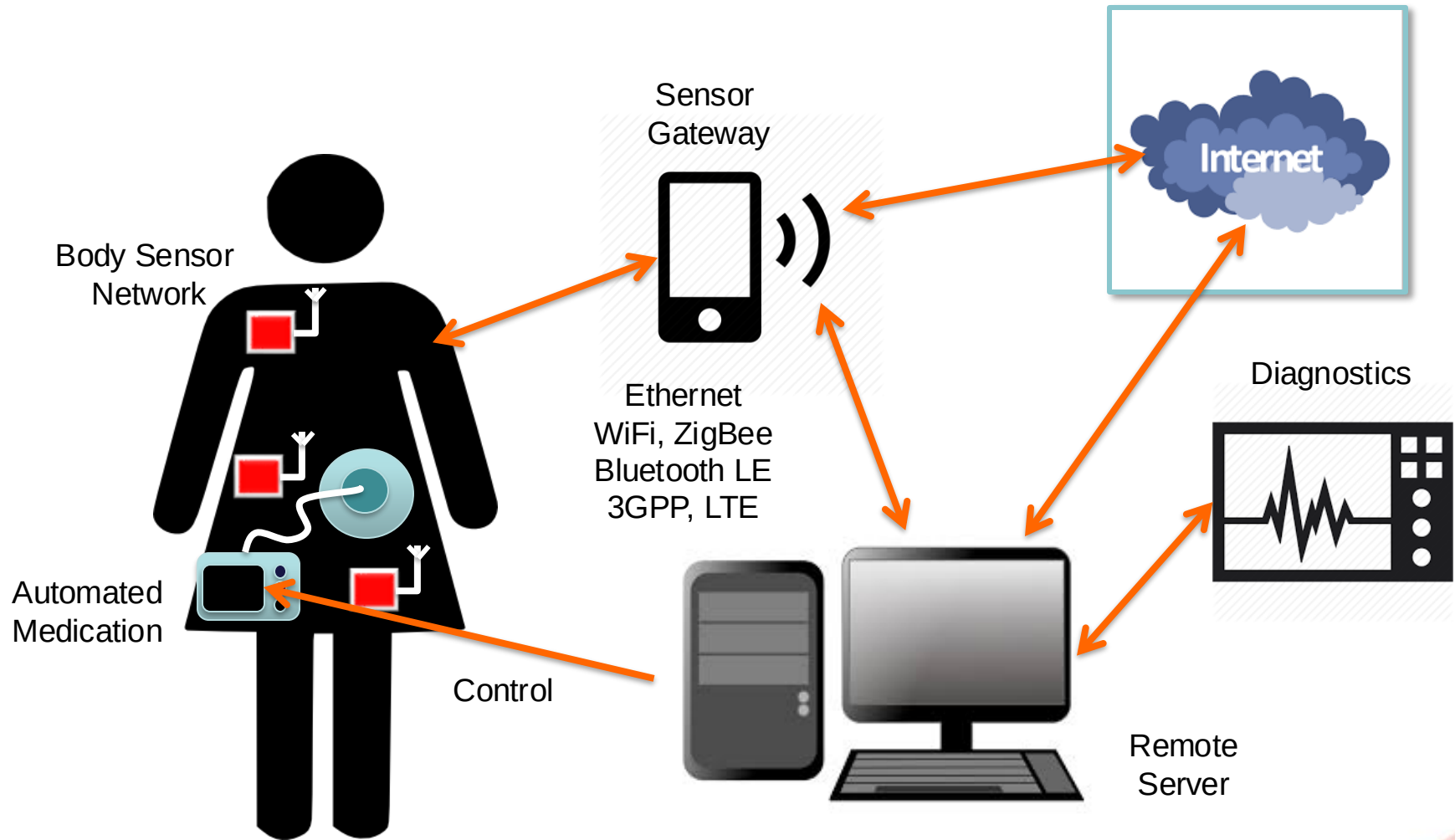


WAREHOUSE OPERATION

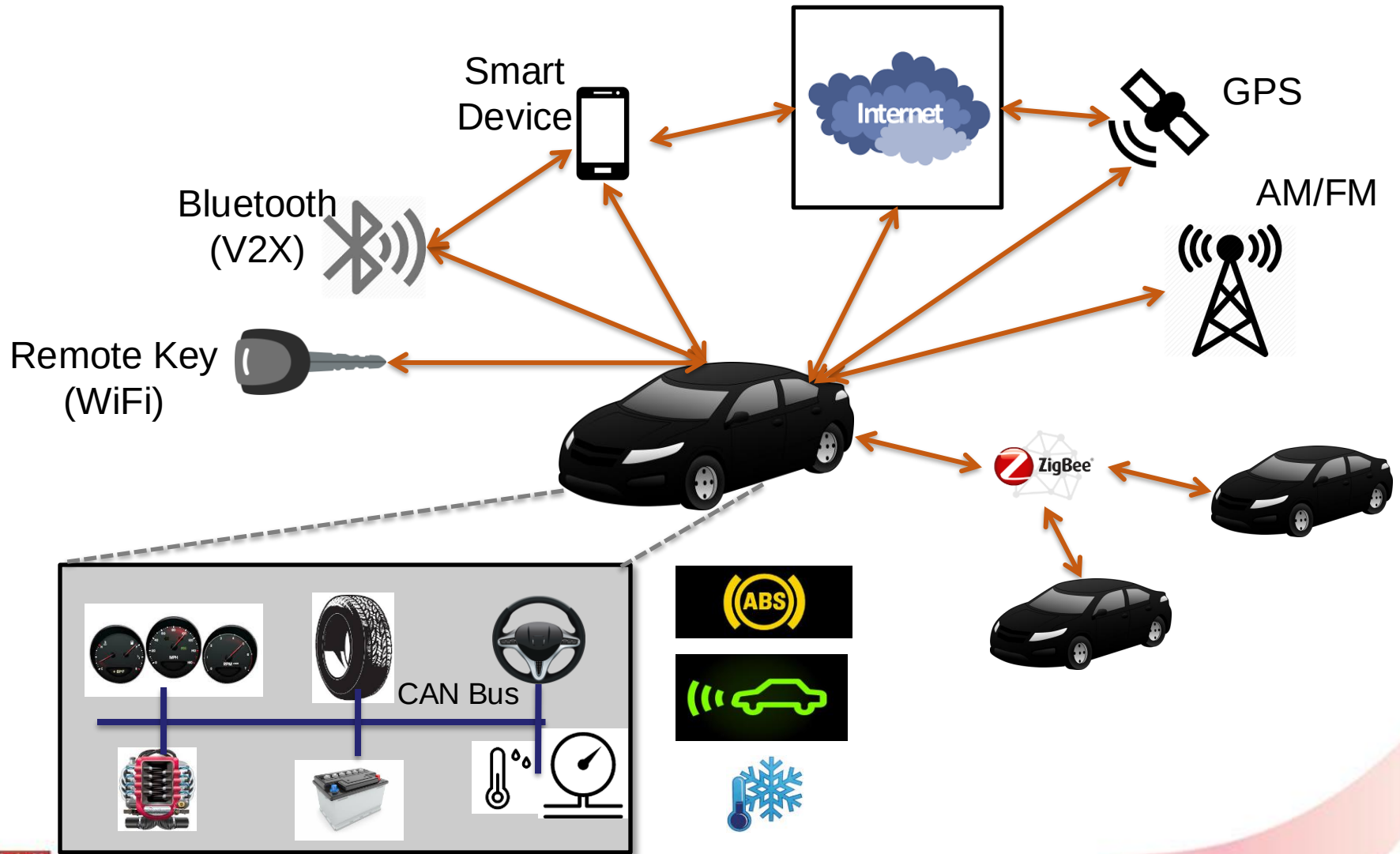




CPS Example: Personalized Healthcare



CPS Example: Automotive



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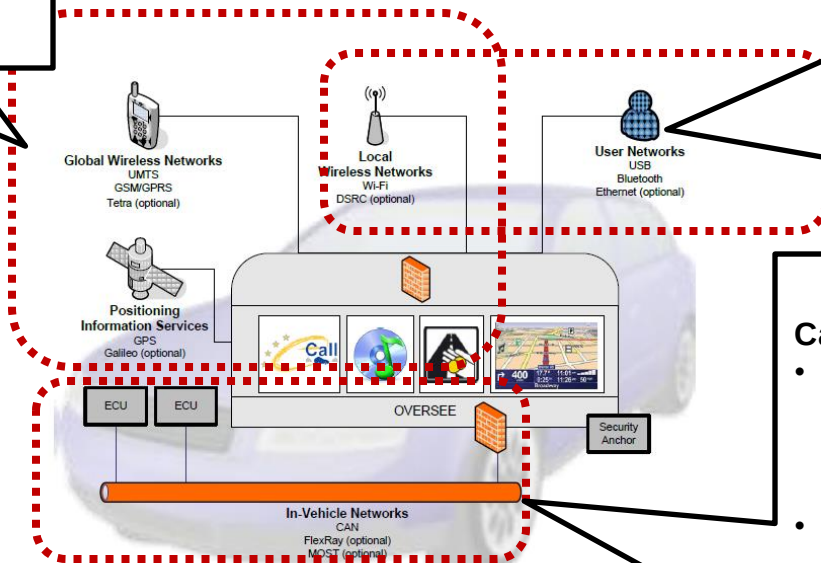
CPS Security: Automotive

Car-to-Cloud

- Security, Privacy
- Identity management
 - public-key protocol
 - Authentication
- Real-time operations

Car-to-X

- Security over Networks
 - Key pre-distribution
- Privacy
 - Untrusted Wireless network
- Information sharing
- Identity management
 - Meet-in-the-middle attack

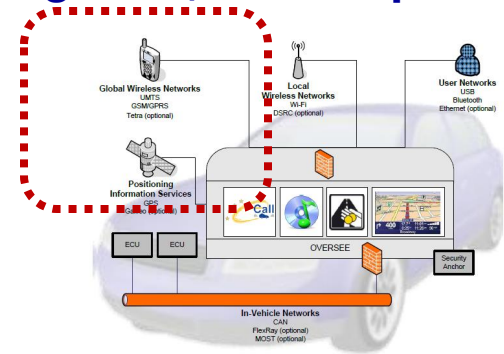


Car Platform

- Security
 - Software stack
 - Storage
 - Network
- Electro-mechanical components
 - Novel attacks
 - Trojans
 - Sensors

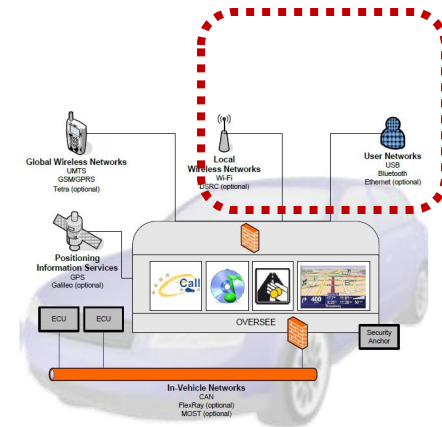
Car-to-Cloud Security

- **AV critically depends on Remote Center for navigation, traffic update**
- **3rd Party Cloud service provider**
 - Data privacy, integrity is vulnerable
 - GPS spoofing
 - Violation of real-time deadlines
 - Identity theft
 - DoS
- **Often reduces problem complexity by offering car-to-car sharing**
 - Malicious car



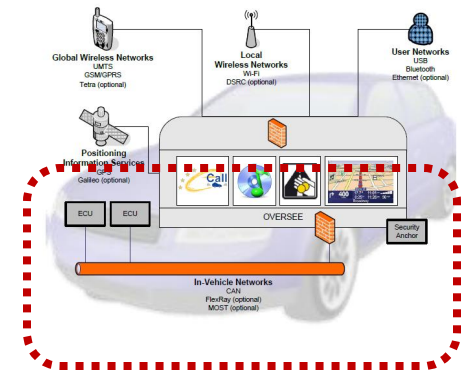
Car-to-X Security

- AV shares information with cars, local user network, traffic system
- Malicious AV/Traffic/User
 - Spoofing attack
 - Data Integrity violation
 - Network jamming
 - LIDAR blocking
 - Meet-in-the-middle
- To be addressed by public-key cryptosystem/distributed key-management techniques



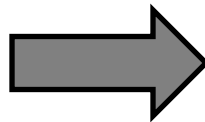
Car Platform Security

- AV is a complex electro-mechanical system
- Remote as well as “*stealthy*” physical attacks possible
 - Protocol violation (*real-time deadlines*)
 - Passive/Active side-channel attacks
 - Replay attacks
 - Software virus
 - Hardware Trojan
- To be addressed by techniques based on ***Root-of-Trust***



Privacy

- Categories of *Individual Privacy*
 - Internal
 - Thoughts/Feelings, Bias, Preferences
 - External
 - Financial, Career, Medical, Ethnicity, Biographical
 - Territorial
 - Tracking, Daily habits, Location visits
 - Social
 - Communication, Family, Friends, Information Dissemination

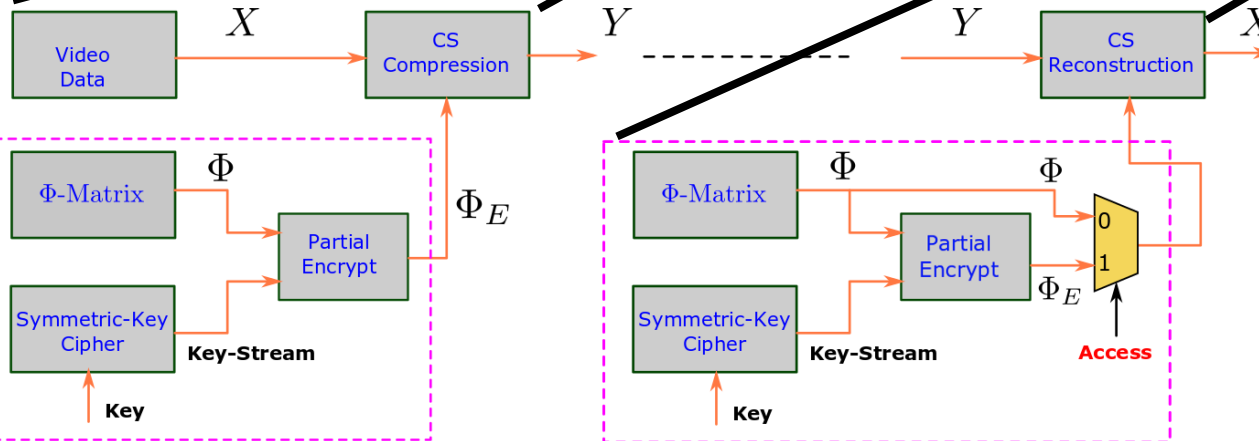
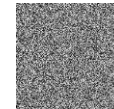


Privacy through Lossy Compression

Camera Module

Raspberry
Pi Board

Laptop



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CPS/IoT Layers

Sensing/Actuation layer

Biological

Optical

Haptic

Acoustic

RFID

Infrared

Thermal

Communication layer

ZigB

VAN

Bluet

NFC

WiMa

BSN

4G, 5G

WAN

WLAN

Ethernet

Modbus

Profibu

CAN

Data Center, Cloud Computing

Application layer

Smart
Traffic Infrastructure

Personalized
Healthcare

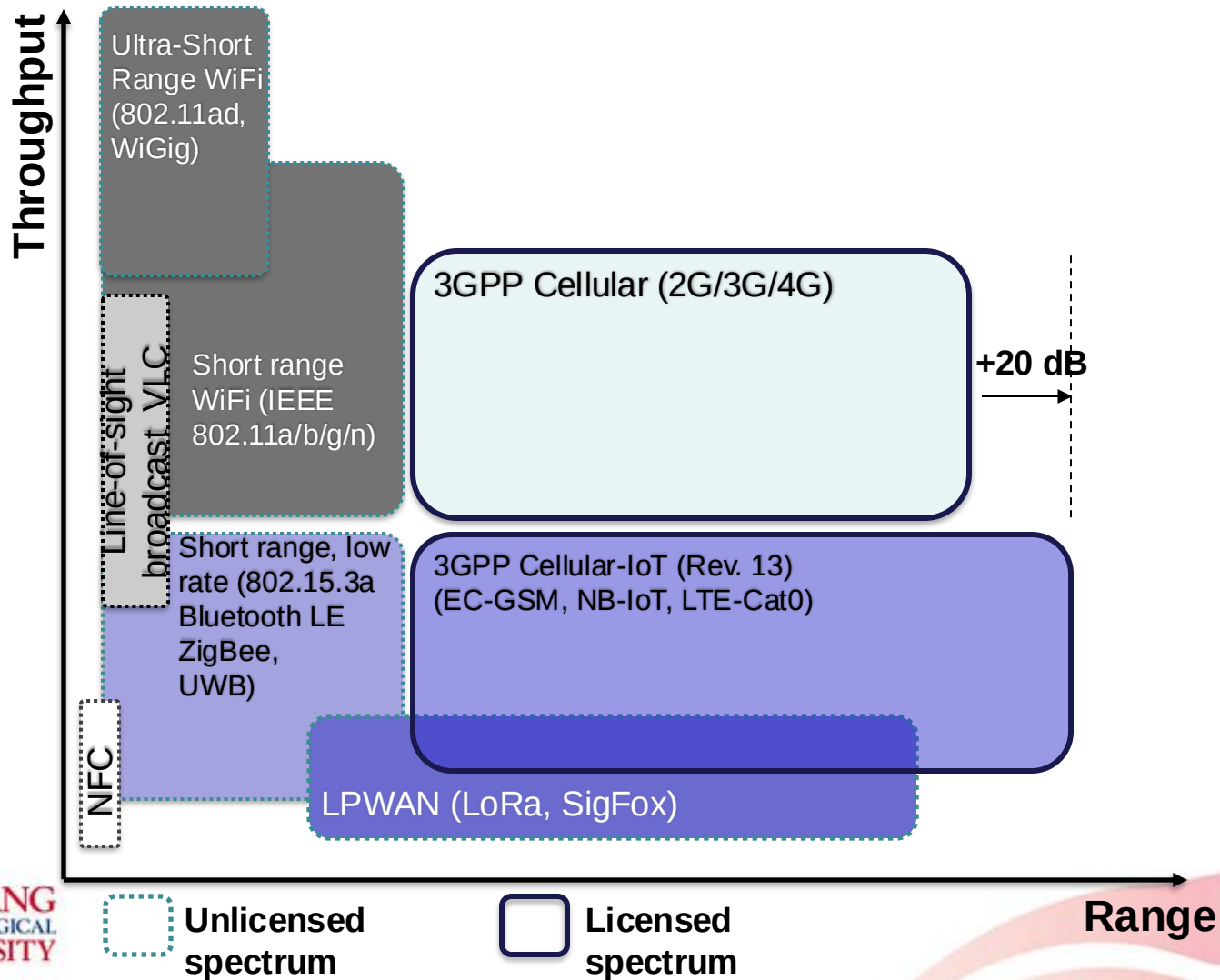
Autonomous
Vehicle

Smart
Grid

Industrial Control
Systems

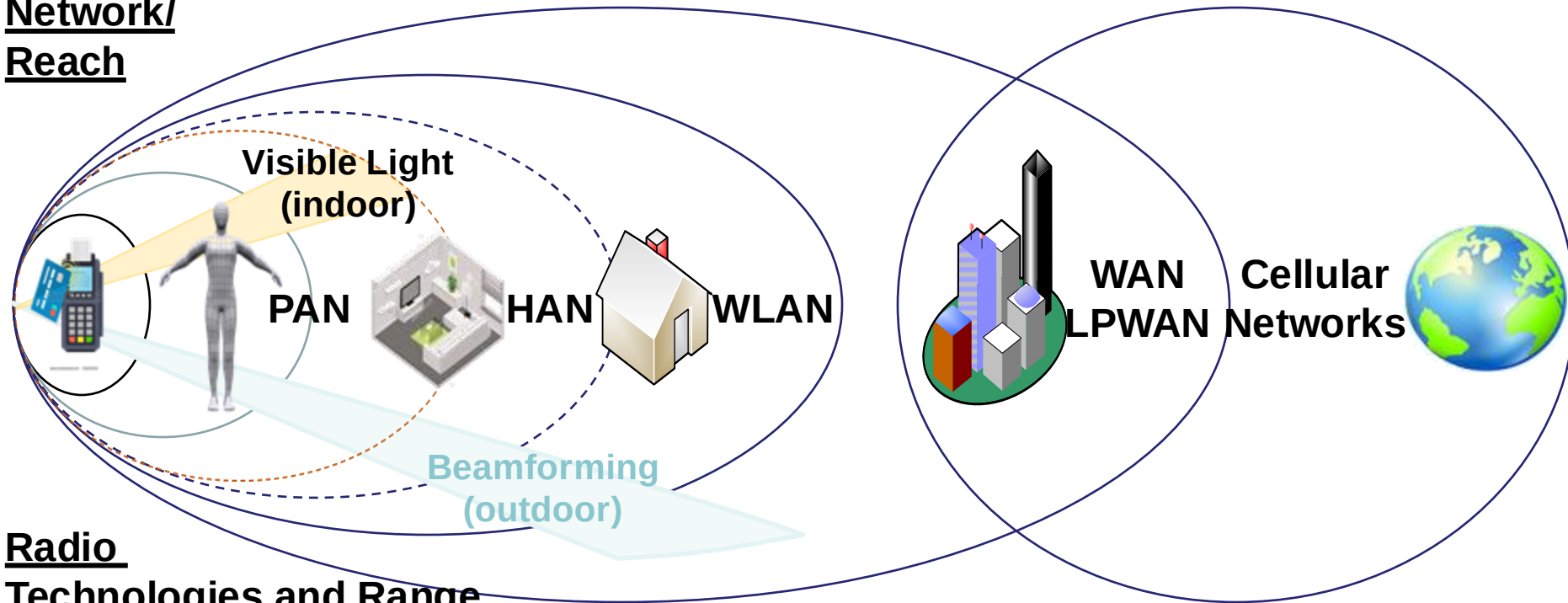
Smart
Manufacturing

CPS/IoT: Spectrum Distribution

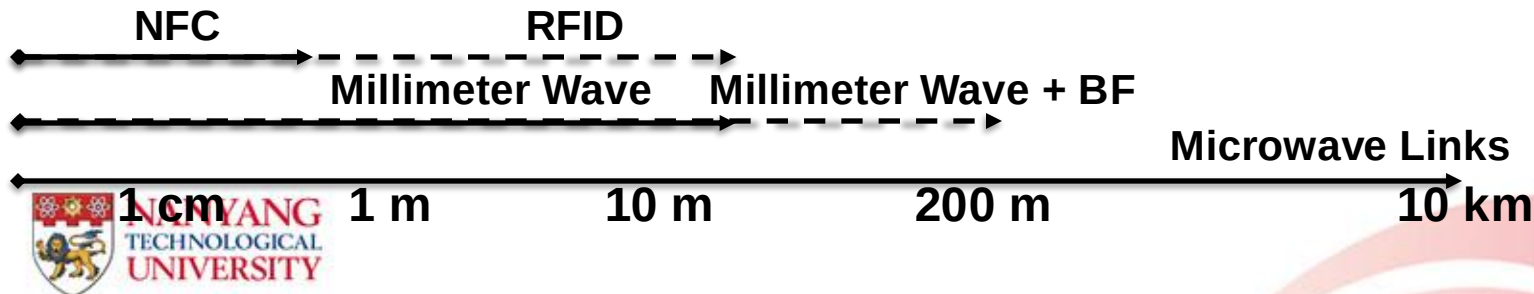


CPS/IoT: Across Standards

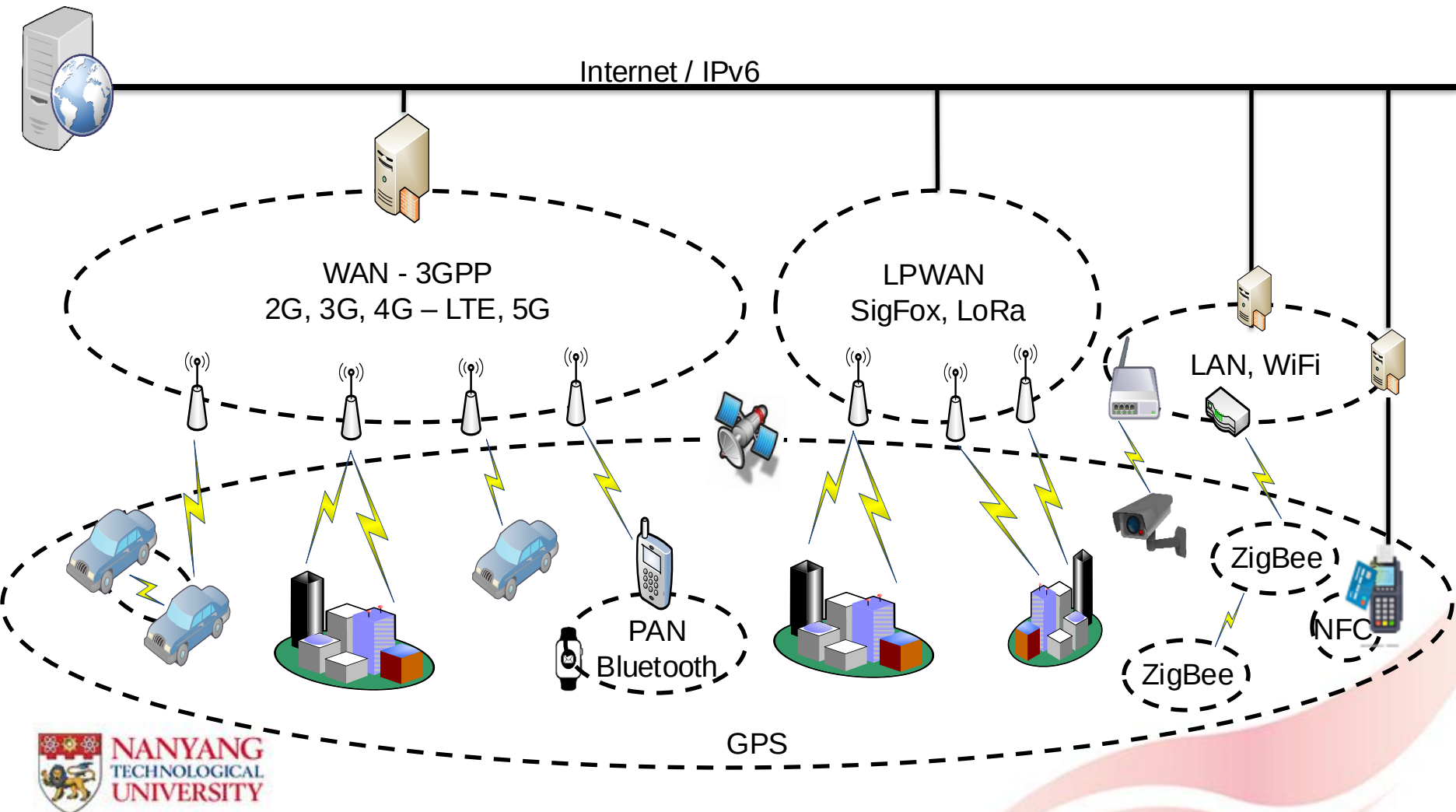
Network/ Reach



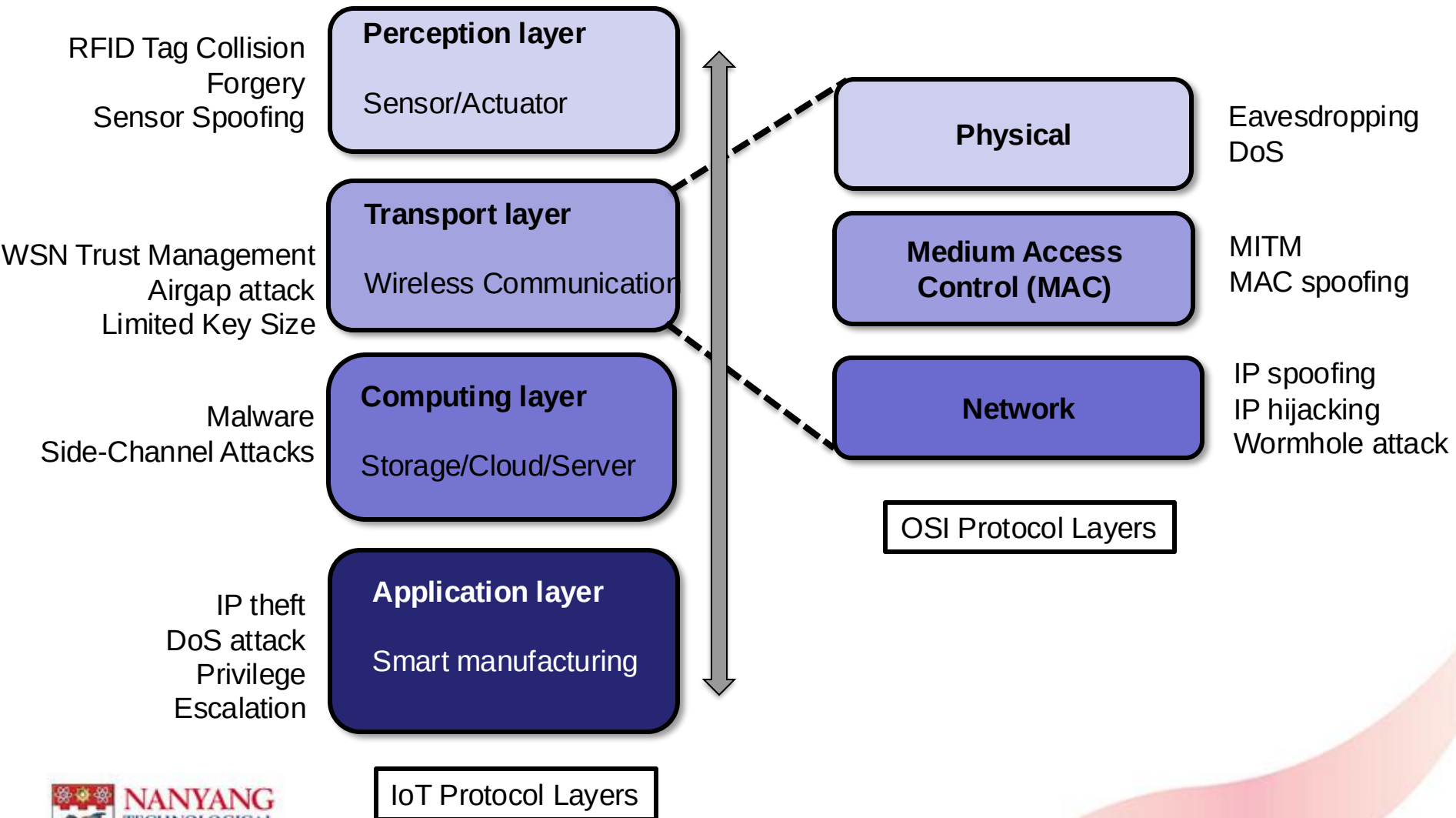
Radio Technologies and Range



CPS/IoT: Heterogeneous Network



Attack Scopes across Layers



Requirements for Wireless Communication in CPS

Property / Requirement	Layer	Description
Network reach	NTW	Area which can be reached within a network itself, i.e., without crossing network boundaries
Radio link range	PHY	Distance that is covered with a single point-to-point link, in a typical application scenario ²
Max. coupling loss	PHY	Maximum signal attenuation (including propagation loss and loss due to obstacles) for reliable reception. Derived from transmit power (transmitter), attenuation (environment), and sensitivity, i.e., minimum received signal strength (receiver)
Peak throughput	PHY	Maximum achievable data rate of a link: often dependent on the actual received signal strength in standards that support multiple data rates
Link traffic load	All	Average amount of traffic generated over a longer time frame (e.g., per day)
Traffic type	All	Traffic pattern of a node: streaming (continuous), bursty (short high throughput), occasional (low rate) bursts
Latency	MAC & NTW	Time required to access the network and to deliver data within the network
Number of devices	All	PHY/MAC: Number of devices that can be present in the same radio link coverage area and access a single point of connection (e.g., access point or base station), NTW: number of devices that can be present in the same wireless network
System capacity	PHY	Overall amount of traffic supported for all nodes (often optimistically related to peak data rate)
Physical security	PHY	Operational technology (OT) security to be provided by the PHY, for example by guarding the sensitive infrastructure
Device power	All	Maximum and average power consumption of a device and its target lifetime
Device complexity	All	Cost and complexity (form factor) of a network node
Network complexity	All	Effort/cost to purchase and deploy a the network infrastructure. Complex networks can only be deployed by operators and are generally used and shared by multiple subscribers which has serious implications on the available services and service guarantees, as well as on the security and access management.
Confidentiality	PHY MAC & NTW	Information theoretic principles are to be used to minimise the information leakage to an eavesdropper. Cryptographic primitives are to be used for encoding the message. Further mechanisms to hide other information leakage, e.g., traffic pattern, routing pattern.
Integrity	MAC & NTW	Message should be accompanied with cryptographic hash to detect tampering
Authenticity	MAC & NTW	Participating nodes should be authenticated, e.g., through key exchange, certification. Messages can be accompanied with digital signatures.
Availability	All	Reliability of a network, including possible PHY/MAC/NTW layer connectivity issues, but also other (e.g., infrastructure related issues, network jamming, DoS)

A. Burg, A. Chattopadhyay and K.-Y. Lam, "Wireless Communication and Security Issues for Cyber-Physical Systems and the Internet-of-Things", *Proceedings of IEEE*, 2018

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➔ *Smart Cards*

- Discussion



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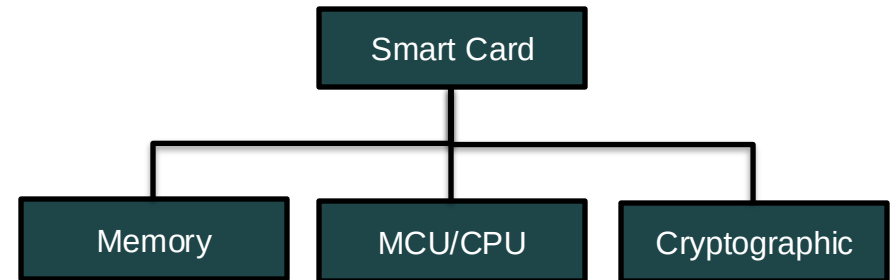
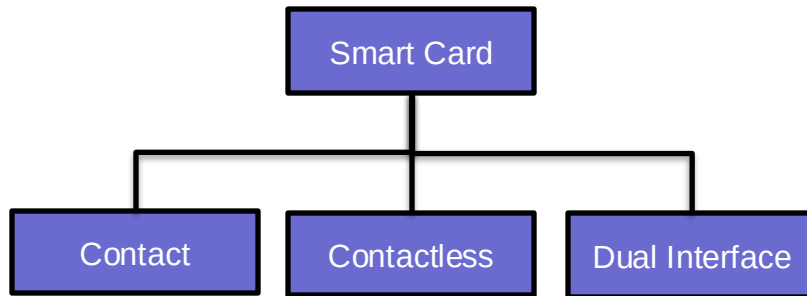
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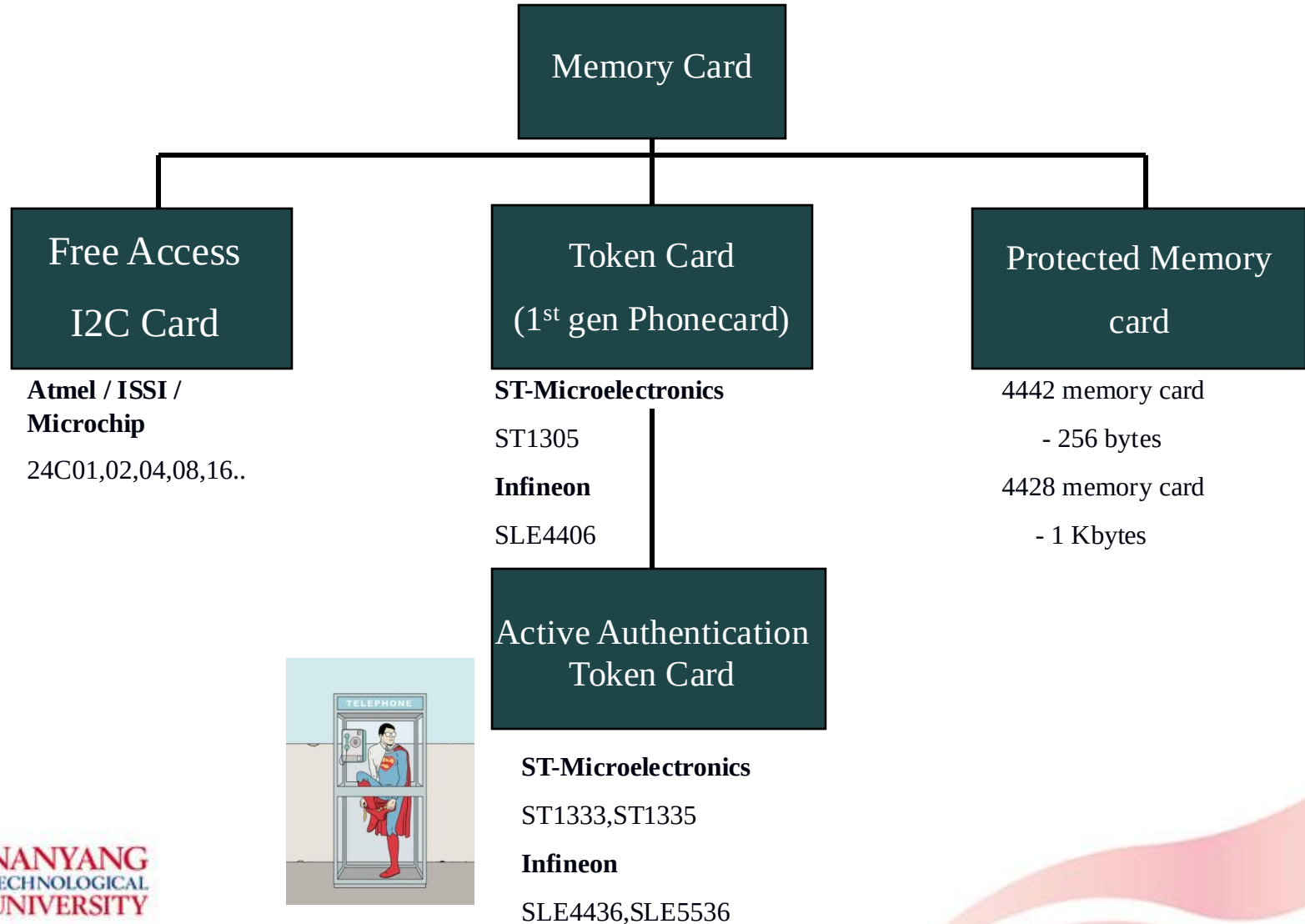


Smart Card / IC Card Family

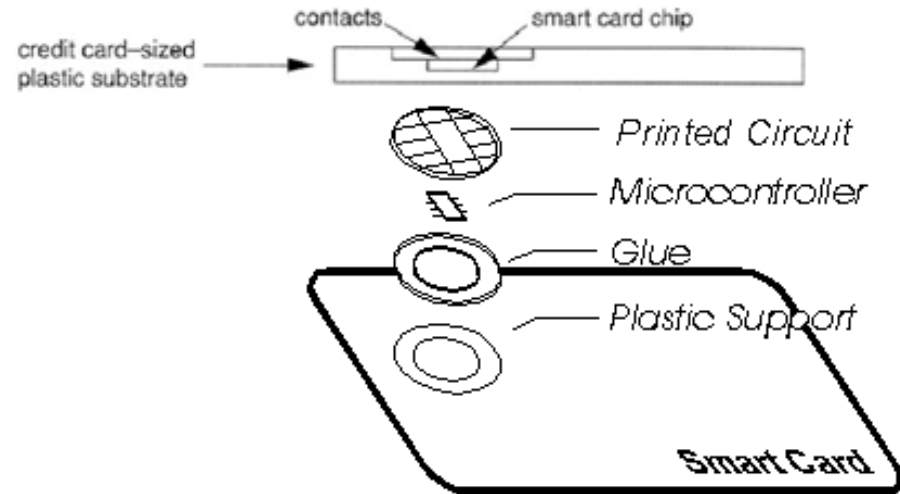
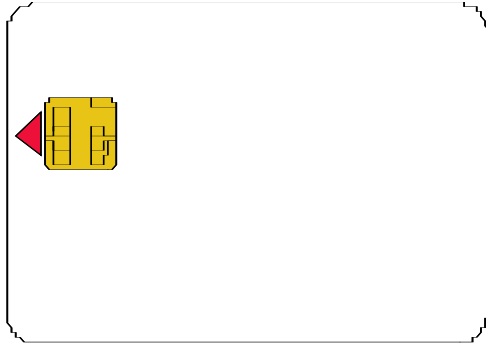


- Contact **Memory** Card
 - Infineon & compatibles
- Contact **CPU** Card
 - GSM SIM, Smart Debit/Credit EMV Card
 - national smart card (banking / ID card)
- Dual Interface Card

Types Of **Contact** Memory Card



What Is A **Contact** Smart Card

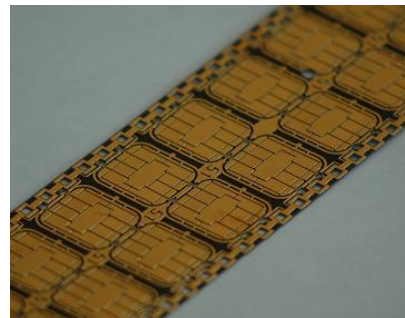
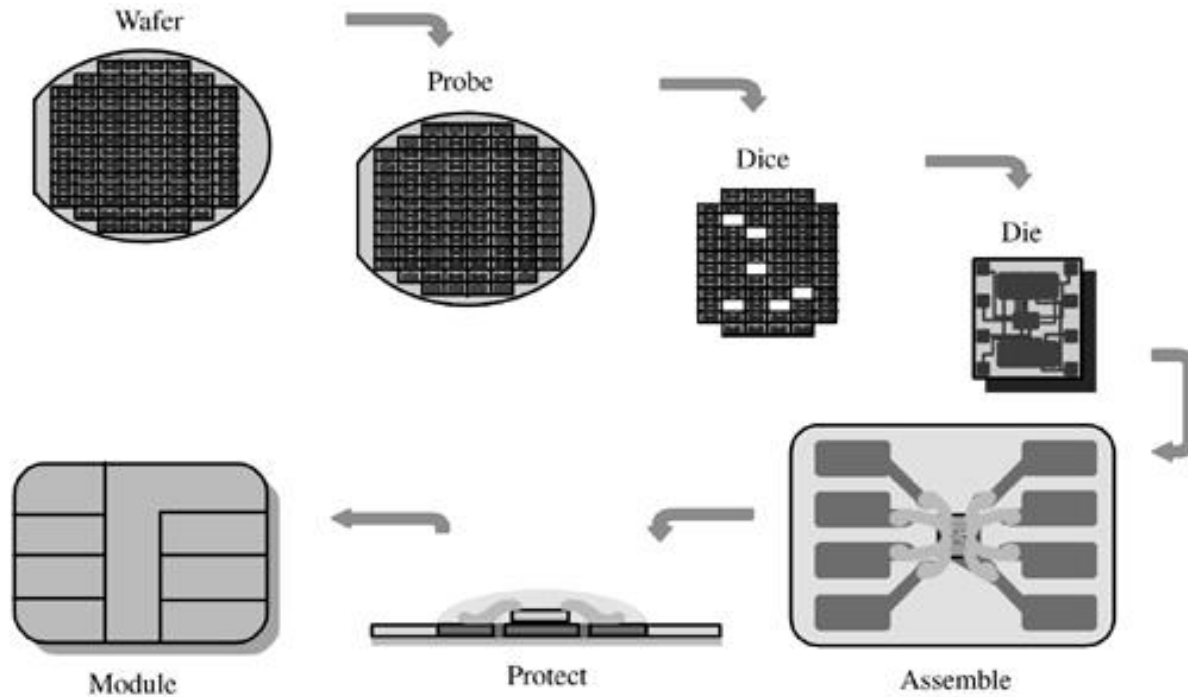


- a credit card size (ID-1) plastic with a single IC chip on board and conforms with ISO-7816
- a smart card comprises of 3 parts
 - contact disc
 - chip
 - plastic body with cavity

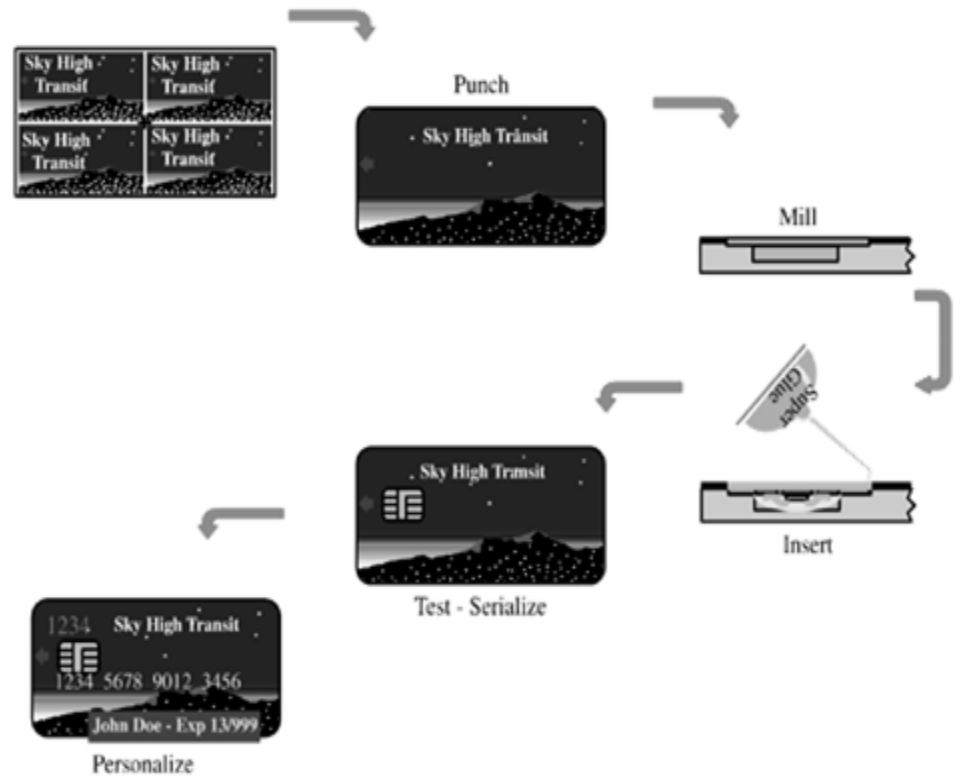
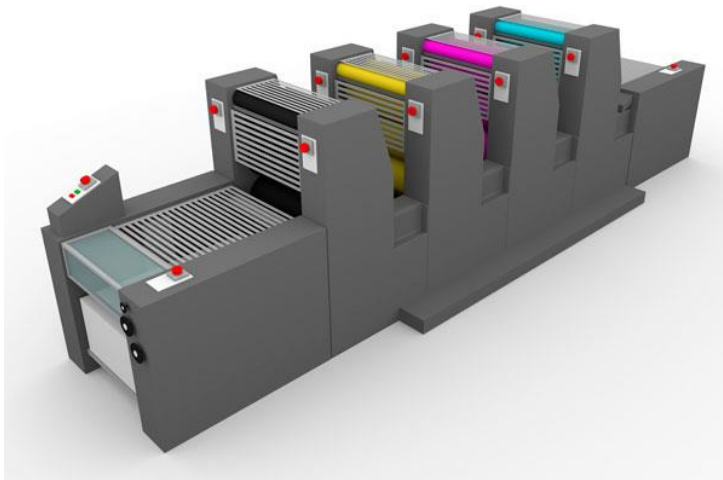
Contact Disc

- Contact disc with the chip is called a micro-module
- 6 or 8 contacts – cost difference of fraction of a cent
- Contact position complies with ISO-7816-2
- *Visually* cannot tell the type of cards from the contact disc
 - Answer-To-Reset says that it is a CPU card

Micro-Module Manufacturing

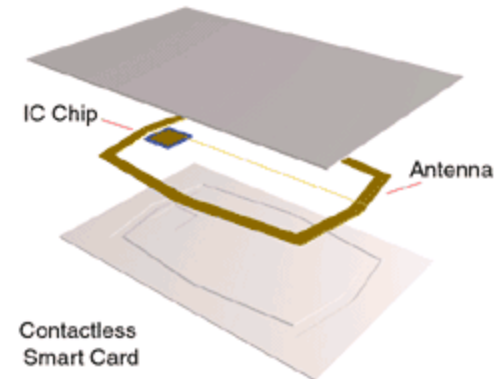


Card Manufacturing

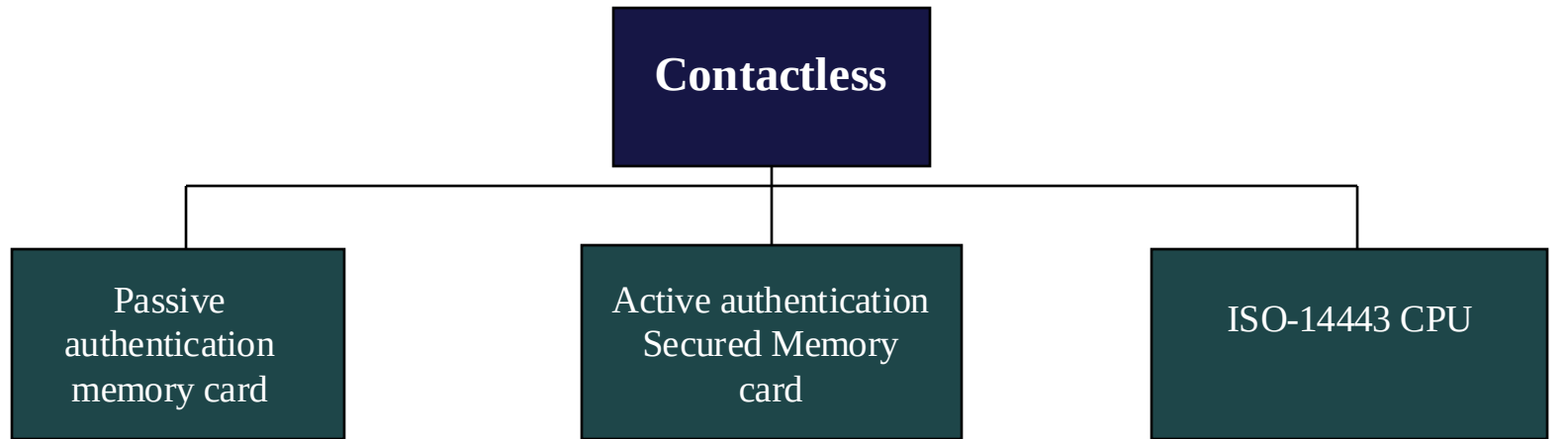


What Is A **Contactless** Smart Card

- Smart card → credit card size card with memory capable of self protection
- If other form factor, referred to as a RF Tag
- Various standards ISO-14443, ISO-15693, ISO-18000-6C (EPC), NFC



Types Of **Contactless** Cards



e.g.

NFC tags from ST-Micro, NXP, Infineon;

Single Trip Ticket (recyclable)

e.g.

Mifare, Mifare compatible

Mifare Ultralight C

e.g.

DESFire,

PBOC 2.0 & 3.0

CEPAS

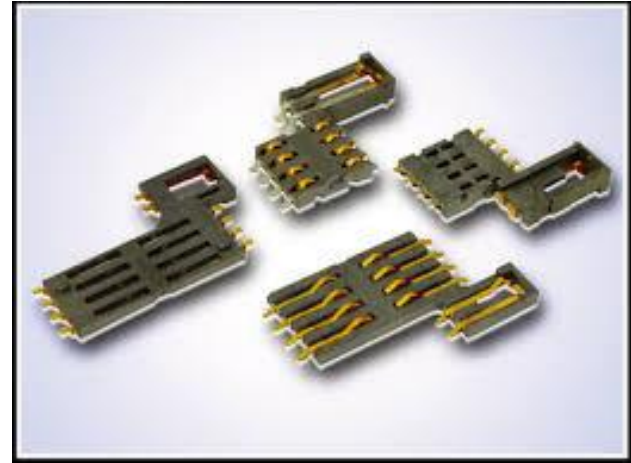
Paypass, Paywave EMV

ICAO passport

Smart Card Acceptors



Card Acceptor



Card Acceptor Contacts



Plug-In Card Acceptor

ISO 14443-1 – Physical Characteristics

This standard defines

- Card dimensions (refer to ISO 7810 for ID-1 cards)
- Surface quality for printing
- Mechanical resistance
- UV and X-ray resistance
- Sensitivity to surrounding magnetic fields

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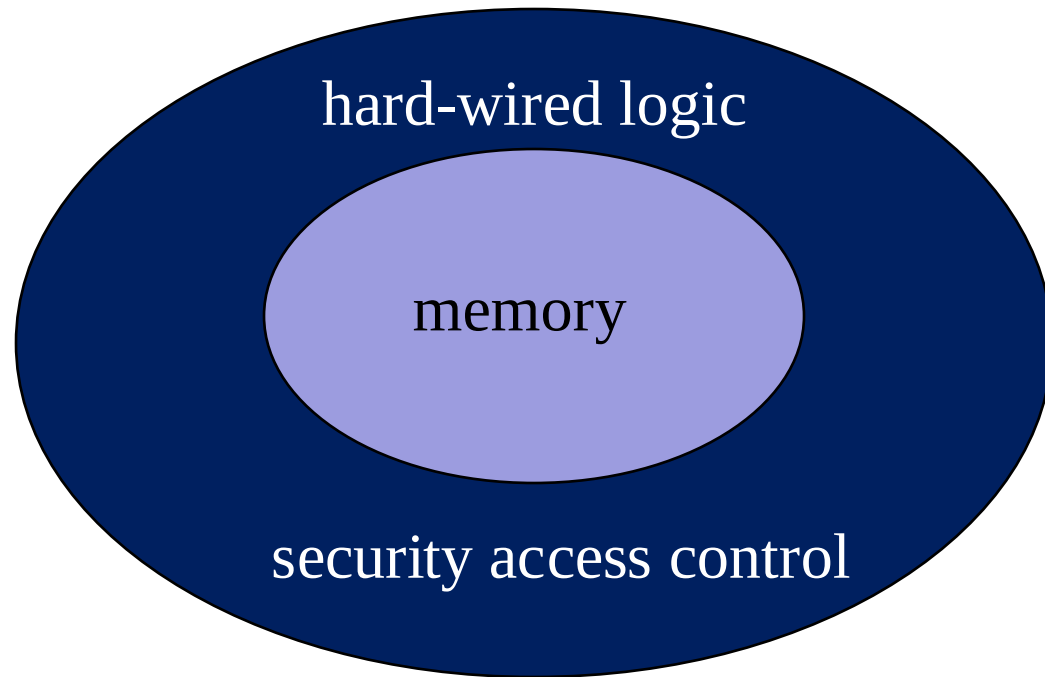


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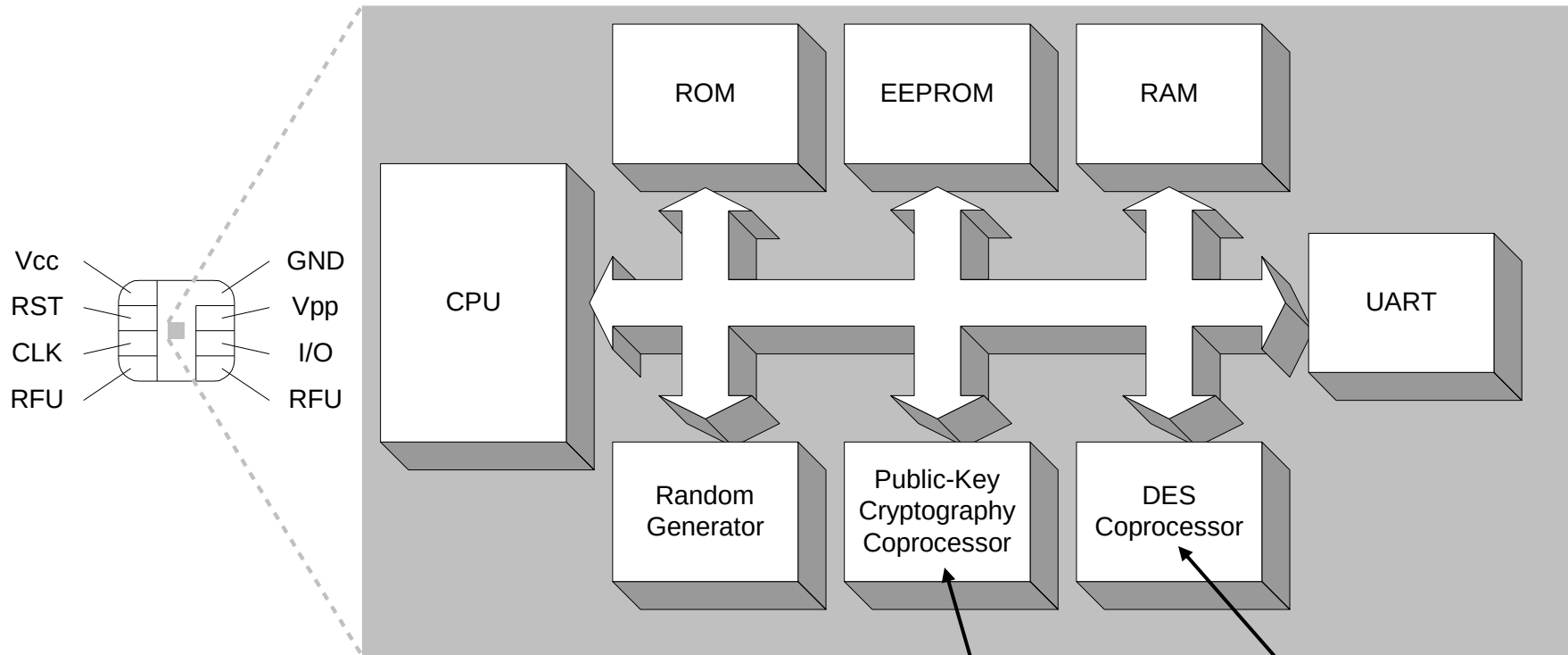
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Memory Card Security Architecture



CPU Card Architecture



DES replaced by AES

(arriving!) Post-Quantum Cryptography

Chip Security

- Goals of attack
 - Extract or clone sensitive data (codes, cryptographic keys)
 - Tamper with s/w execution, bypass security mechanism, modify data
- Typical Security Features of smart card chip
 - Chip hardware
 - Chip OS
- Essential that the application system security design fully exploits the security features and patch up any security limitations and weaknesses, if any

Physical Attacks, Countermeasures

- Physical or invasive attack methods
 - Chip modification, signal probing, memory content or secret retrieval
 - Fault injection – power supply disturbance, clock frequency, temperature, light, laser
- Physical / invasive attack countermeasures
 - Security sensors – voltage, frequency, passivation layer, power supply glitch sensors
 - Memory scrambling, memory encryption, bus encryption, glue logic layout, dynamic shield covering entire die surface, EEPROM 2 page destroy mechanism
 - RAM integrity check, EEPROM double READ, Code integrity check for critical routine

Side Channel Attacks, Countermeasures

- Taking advantage of side channel information that leaks secret
 - Timing attack, power consumption, magnetic, electrical analysis
- Countermeasures against side-channel attack
 - Hardware internal jittered clock mode to add random delays
 - Software adding random dummy clocks
 - Scrambling, Encryption

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Secure Printing

Main Goal of Secure/Security Printing

Prevent forgery or counterfeiting

Typical Application

National ID cards, passport, banking card

Typical Examples

- UV Printing
- Microtext
- Multiple laser image (MLI)
- Watermark
- Hologram



Secure Printing



Secure Printing



(Source: HK Immigration Department)

Secure Printing – *more examples*

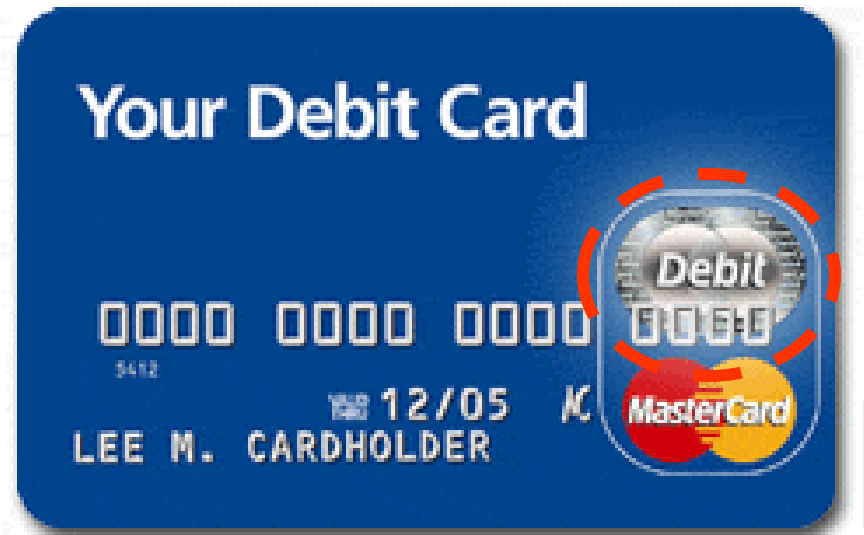


HOLOGRAM



More about hologram

Setup charge for molding may cost
US \$2,000!



Secure Printing – *more examples*



Microtext

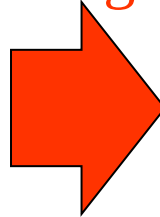


Secure Printing – *more examples*

UV Printing



UV light



Secure Printing – *more examples*



Watermark



Glossary (for Secure Printing)

Guilloche: printed security lines, where the layout of intersections and geometry are unique.

Hologram: Is a unique form of photographic printing that is flat optical image to the naked eyes and provides a three-dimensional effect on a flat surface.

Microtext: This involves extremely small text which is small enough to be indiscernible to the naked eye.

Multiple Laser Image (MLI): Multiple laser image can be viewed at different angels.

Optical Variable Ink (OVI): a high security feature showing different colors as the angle of view changes.

UV (Ultraviolet) Printing: is invisible under regular illumination. By viewing the text/graphic under UV light, they become visible in yellow color.

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RF ID

LF

Low Frequency

- 125-134 KHz
- Small memory (few tens byte)
- No need for inter-operability
- Low security
- Animal tagging

HF

High Frequency

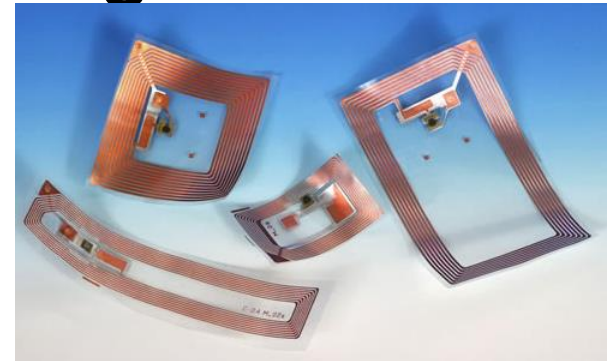
- 13.56 MHz
- ISO14443, ISO15693, NFC
- Various memory size from few tens bytes to several tens Kbytes
- Low, Medium, High Security
- NFC tags
- E-purse, EMV, Passport, Organization / National ID card

UHF

Ultra High Frequency

- 856-960 MHz
- ISO 18000-6C
- Detection distance of up to 10+ m
- Small memory
- inter-operability
- Low, medium security
- Logistics, retailer, road toll, airport luggage tracking

Different Forms Of RF ID Tags



What is NFC Tag

Property	Type 1	Type 2	Type 3	Type 4	Type 5
Standard	ISO/IEC 14443A	ISO/IEC 14443A	ISO/IEC 18092 JIS X 6319-4 FELICA	ISO/IEC 14443A ISO/IEC 14443B	ISO/IEC 15693
Memory	96 bytes to 2 Kbytes	48 bytes to 2 Kbytes	2 Kbytes	32 Kbytes	64 Kbytes
Data rate	106 kbit/s	106 kbit/s	212 kbit/s, 424 kbit/s	106 kbit/s, 212 kbit/s, 424 kbit/s	26.48 kbit/s
Capability	Read Re-write Read-only	Read Re-write Read-only	Read Re-write Read-only	Read Re-write Read-only Factory- configured	Read Re-write Read-only
Anti-collision	No	Yes	Yes	Yes	Yes
Notes	Simple, cost effective	-	Higher cost, complex applications	-	Vicinity area

e.g. Topaz

e.g. My-D, NTag

e.g. Felica

e.g. ST25TA, ST25TB

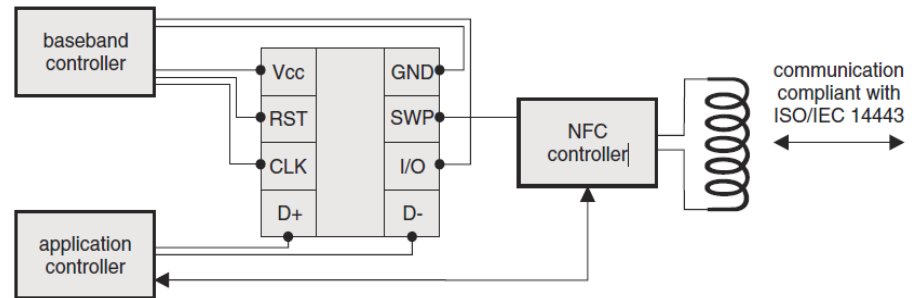
e.g. ST25TV



Source: ST Microelectronics NFC Guide

https://www.st.com/content/ccc/resource/technical/document/technical_note/f9/a8/5a/0f/61/bf/42/29/DM00190233.pdf/files/DM00190233.pdf/jcr:content/translations/en.DM00190233.pdf

Near Field Communication (NFC)



- Inspired by RF ID to be used in mobile phone in 2002.
- In 2004 NXP, Sony and Nokia founded the NFC Forum.
- An **NFC phone** works in 3 modes:
 - Card Emulation Mode
 - Reader/Writer Mode
 - Peer-To-Peer Mode
- Driven by an application (applet) installed in Secured Element (SE), as part of SIM (operator) or as part of eSE (mobile device)

Trusted Service Manager (TSM)

- Trusted Service Manager (TSM) is a unifying service provider that enables various modes for the NFC application operator
- NFC application operator needs a card emulation applet inside a Secured Element (SE) (NFC SIM, eSE)
 - **NFC SIM**: Need to work with different telecom operators (some may want exclusivity)
 - **eSE**: Need to work with different mobile manufacturers
- Need to qualify different phone models as performance can differ
 - To qualify the card emulation application working with mobile
 - To be securely loaded into the SE

EZ-Link and SimplyGo

- EZ-link started in 2001, with a card-based wallet requiring top-ups. This is now upgraded to *account-linked* system in 2021.
- SimplyGo started in 2019, with *account-linked* (e.g. Mastercard) payment.
- Follows Specification for Contactless e-Purse Application (CEPAS).
- Account-linked payment systems in smartphone, it utilizes *Trusted Service Manager (TSM)* to maintain the wallet.
- EZ-link used Triple DES^[1], which is deprecated by NIST since a major vulnerability was discovered^[2]. Other smartcards (e.g. Mifare) using Triple DES has been shown to be vulnerable^[3]. Privacy leaks of EZ-link has been studied earlier with side-channel attack setting in 2014^[4].

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What did we learn?

- **What are various CPS/IoT ?**
 - Few Examples
 - Attack Classification Example → highlight smart automotive
 - Communication Protocols → heterogeneity
- **Smart cards?**
 - Chip security
 - Secure printing (card protection)
 - NFC/RFID (communication protection)
 - Trusted Service Manager (implementation)



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Further Readings

Automata and *Examples*

Automata theory

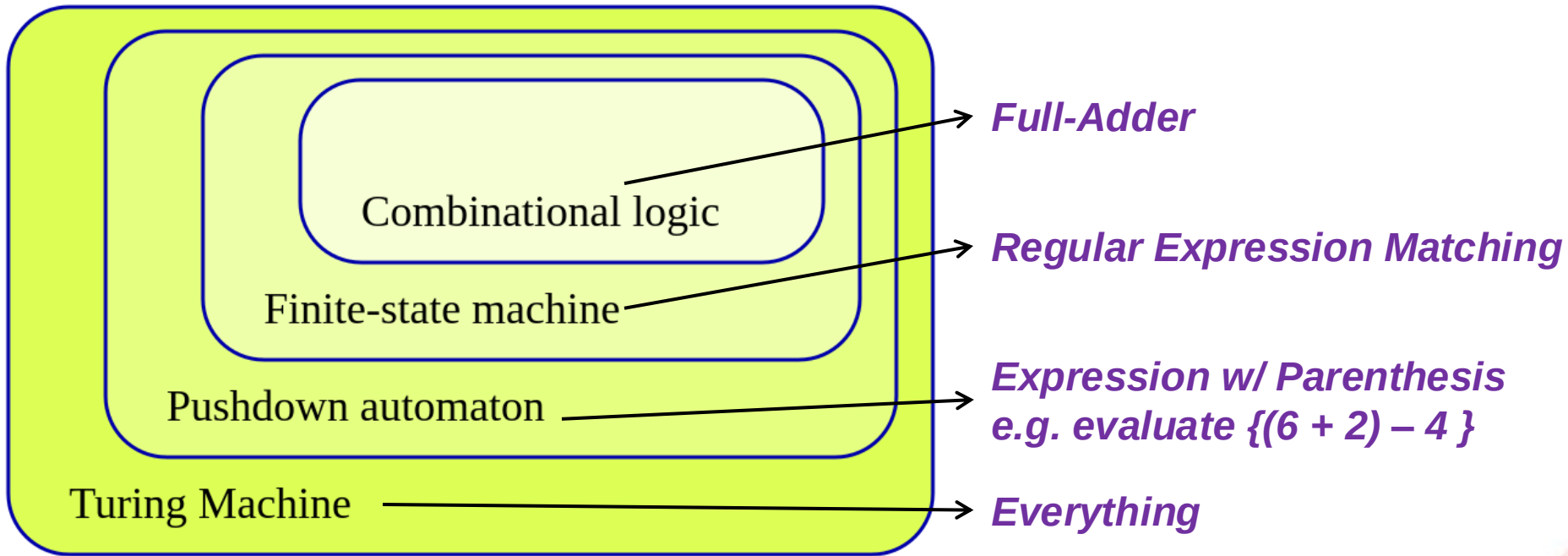


Image source: wikipedia

Trusted Service Manager

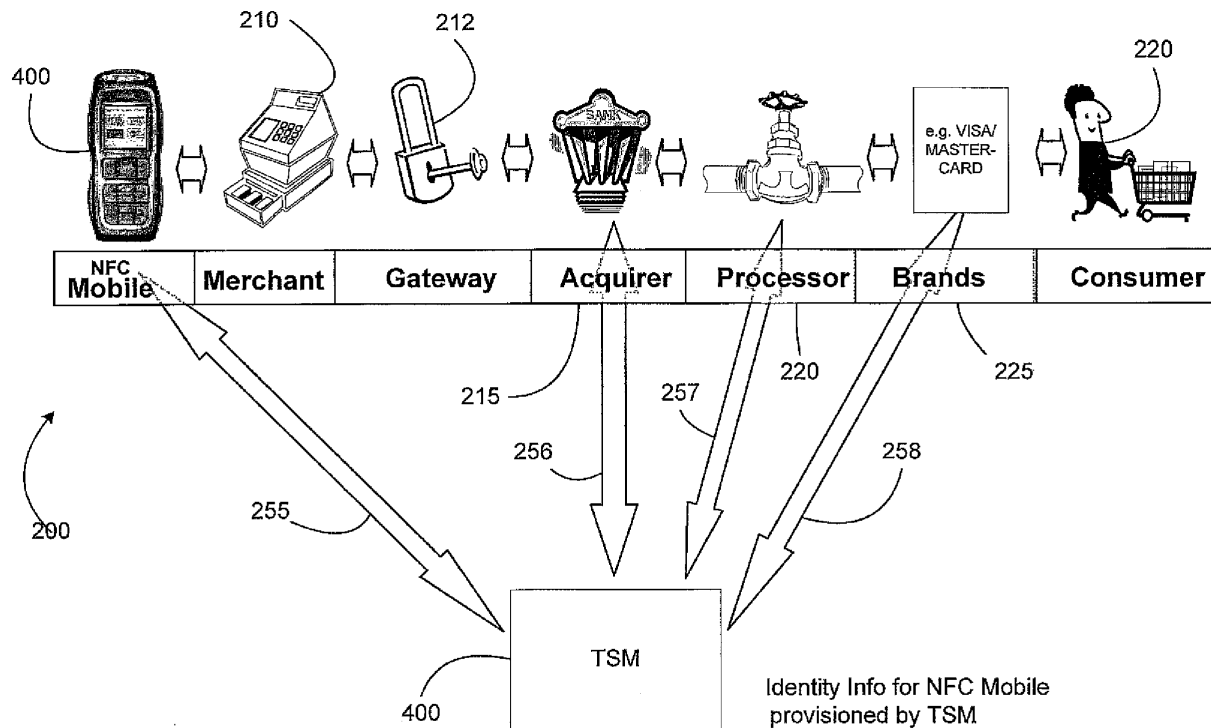


FIG. 2

Trusted Service Manager

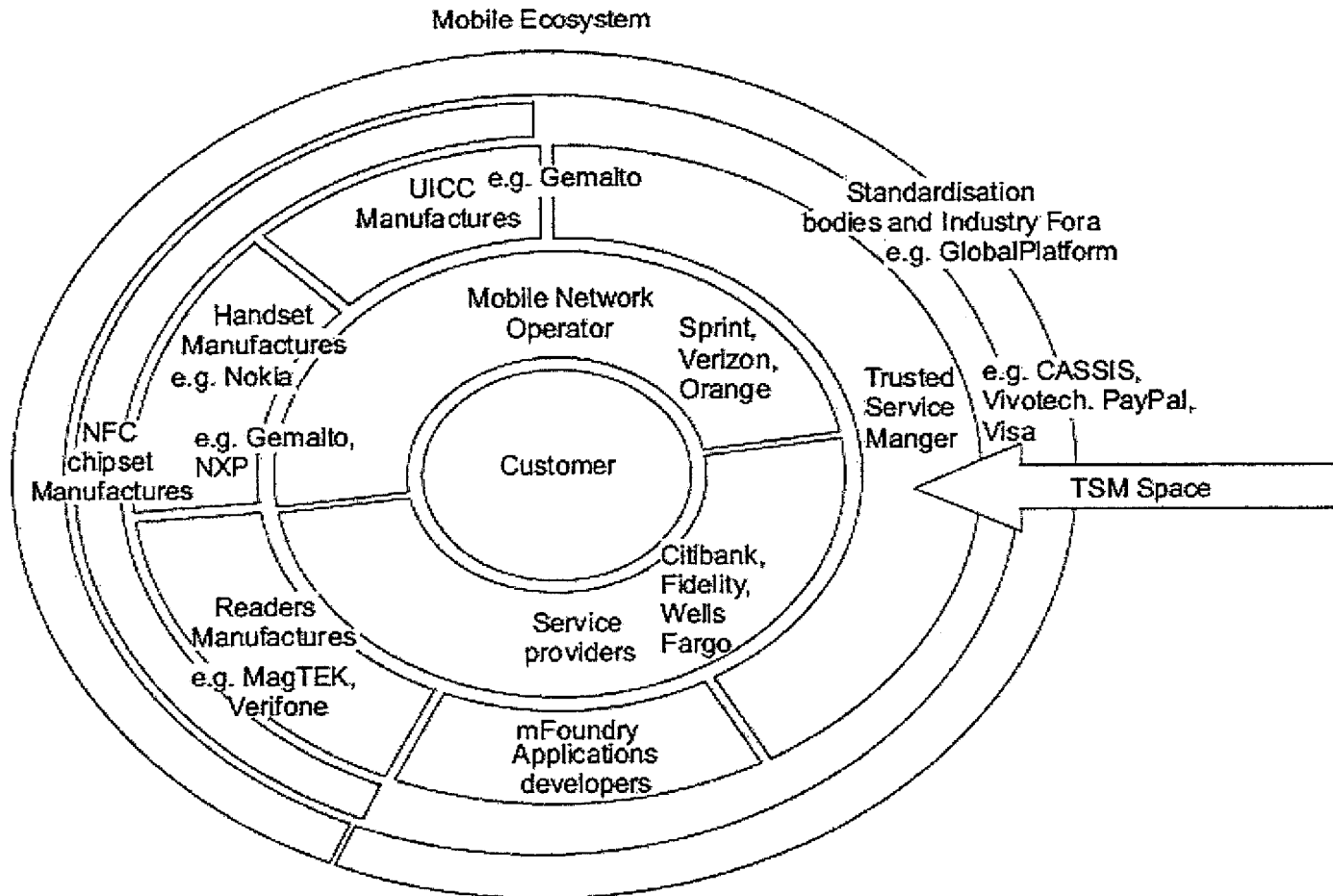


FIG. 1

Trusted Service Manager: Tasks

- User registration: Corresponding to a bank
- Managing Digital Certificates: User identity
- Authentication and Verification: Transactions of the mobile wallet with a payment terminal
- The above services are delivered using the underlying platform, in particular
 - **SE**: Storage of keys, passwords, identity
 - **Secure Communication**: NFC
 - **TEE**: Secure Microprocessor for providing security services (e.g., encryption, authentication)

The End